Arithmetic Mental Task of EEG Signal Classification Using Statistical Modeling of The DWT Coefficient

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***Abstract*— This assertion proposed a statistical modeling of recorded EEG signal where Gaussian distribution is utilized to classify the signals from a Mental Arithmetic Task for categorizing Bad counters and Good-counters (those who required excessive efforts for the task or not respectively) of 36 subjects. Bad-counters finish the arithmetic task with a high level of difficulty and Good-counters finish the arithmetic task with no difficulty. The dataset of this paper carries EEG recordings of 36 subject's performance of mental arithmetic tasks in the active and resting state. The dataset used here is taken from PhysioNet database platform where 22 channels are used. Here Discrete wavelet transform is used to divide the dataset signals into a number of sets. The main aspects of this paper are to set up a model which can analyze arithmetic Task signals to find Good or Bad it is. As a result of this work, after completing the classification of EEG data, proposed statistical modeling yielded a 72.22% accuracy.**

Keywords*— *Electroencephalogram (EEG), Discrete wavelet transform (DWT), Channel selection, Mental arithmetic task, Gaussian distribution.

I. Introduction

Brain is an organ of electro-chemicals and an electroencephalogram (EEG) signal can be used for recording its electrical activity. The EEG signals are extremely erratic in character and it provides significant knowledge regarding the condition of the brain. They are non-Gaussian, non-linear and non-stationary by definition. EEG signals are used for many applications for example: diagnosis of seizure, epilepsy, sleep disorders and detection of tumors etc. It is also used for mental task classification [1].

A standard EEG signal has 10-100v amplitude and 1-100Hz frequency while measured from the scalp. Electrodes are inserted on the subject's scalp to determine the signals. The 10–20 electrode placement scheme developed for Electroencephalography specifies electrode positions. The relationship between the fundamental area of the cerebral cortex and the location of an electrode created the 10-20 system of an EEG. The frontal, paternal, temporal, and occipital lobes are located in the cerebral cortex and these four cortices' electrical activity are captured by EEG signals, which aid in understanding their function [2].

EEG recording used in a wide range of applications, starting from basic research on brain organization, development, and cognition to clinical research and claims for managing patient illness conditions. Some applications are Brain Computer Interface, Epilepsy monitoring, Sleep studies, Produce biofeedback, drug and surgical treatments. We worked on the Brain Computer Interface in this assertion, which is a route for communicating directly between the human nervous system and output devices.

Fourier Transform (FT), Fast Fourier Transform (FFT), Short Time Fourier Transform (STFT) and Wavelet Transform (WT) are the mathematical tools which are introduced for EEG signal feature extraction. Wavelet transform (WT) offers a broad range of applications in the area of Digital Signal Processing. Wavelet transforms are classified as continuous, discrete. And it's a combined use of the process of convolution, which involves reversing, shifting, multiplying, and integrating them [3].

Since the EEGs discovery, people have postulated that it may be used for communication and control, allowing the brain to perform on the situation without the usual middlemen of peripheral nerves and muscles. Brain computer interface's purpose is to create a new type of communication system that allows the human brain to control equipment and devices without the usage of muscle movements. BCI uses machine learning to transform EEG signals by brain activity into control directives. BCI is a biological communication system that connects the exterior world within real time also in a natural way. By converting brain signals into control commands BCI accelerates the communication between the brain and external world devices. Those who are suffering a neuromuscular disease, BCI can be highly benefited by using their brain signal to access multiple tasks. BCI working principle consists of five steps- first step is measurement brain activity, second is signal preprocessing, third step is extraction of features, fourth step is classification, fifth step is control interface which converts the classified signals to readable data for devices such as computers. Those phases are a very important part that highly influence the accuracy of the proposed BCI model.

Mental cognitive workload refers to the mental effort level required to complete different tasks such as mental arithmetic problem solving, reading or writing and they can be evaluated by tracking changes in neurophysiological parameters like EEGs. It indicated the amount of mental effort exerted by a task on an individual. Identification of mental workload during cognitive tasks is an important area in cognitive neuroscience and other real-world applications. Cognitive workload measurement is an essential problem in numerous study and application regions of human computer interaction because it may be used to assess the potency of interfaces and create adjustable interaction techniques. The advancement of real-time measure of the mental workload may lead to the formation of intelligent systems which can adapt instruction and maintain subjects in an ideal

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variety of mental workload to make sure learning ability, or even inform subjects regarding their stress levels throughout these workload. [4-7].

Serial subtraction of two numbers is one of the Arithmetic tasks in this study. When the EEGs were recorded the participants sat comfortably in a dark soundproof chamber. Participants were instructed to count silently and not to speak or not to use finger movements. After 3 minutes of acclimating to the laboratory systems, the EEG was recorded in the rest state with closed eyelids for the next 3 minutes. The participants next completed a 4-minute mental arithmetic task.

The earlier research that has been done in the domain of Mental arithmetic task classification using LSTM architecture method by B. Ganguly [8] and another work where the Fourier Decomposition Method is used for the classification of Mental tasks by B. Fatimah [9]. And authors in [1] considered classification accuracy of automatic detection of metal arithmetic tasks using EEG signals.

The aim of this assertion is to capture the electrical activity of the brain using EEG signals collected from humans performing a mental cognitive workload task (mental arithmetic task). We used wavelet transform to preprocess our EEG dataset and then proposed a statistical model for the DWT coefficients of EEG signals. Using those recovered modeling parameters, SVM and KNN classifiers were utilized to classify Arithmetic mental task signals. We have checked that our chosen statistical model is appropriate for our data by using two parameters which are- Bhattacharyya Distance (BD) and Geometrical Separability Index (GSI).

II. DATASET

In our assertions, here we used a dataset collected from the open-source public PhysioNet database. Dataset presents recordings from the subjects performing mental task-serial substructions following the International 10/20 system. Here 36 healthy subjects participated during the collection of EEG recordings with the aim of investigating EEG match up of mental cognitive workload, specifically arithmetic mental tasks. The EEGs are recorded using a 23 channel Neurocom monopolar device. The electrodes made of silver or silver chloride are placed on the subject's scalp at symmetrical anterior frontal (Fp1, Fp2), frontal (F3, F4, Fz, F7, F8), central (C3, C4, Cz), parietal (P3, P4, Pz), occipital (O1, O2), and temporal (T3, T4, T5, T6) recording locations as per 10/20 International system. The inter-electrode impedance was less than 5 kΩ. The sampling frequency for each channel was 500 Hz. A high-pass filter with a cut-off frequency of 0.5 Hz, a low-pass filter with a cut-off frequency of 45 Hz, and a power line notch filter with a cut-off frequency of 50 Hz were utilized where the amplification tract's time constant was 0.3 s. In every recording 60 seconds for mental workload counting and 180 seconds for resting state included EEG segments with no artifacts. In the recorded sample, a total of 36 subjects were included. There are two different subsets in this dataset, they are named Bad counters and Good-counters (those who required excessive efforts for the task or not respectively). Bad-counters finish the arithmetic task with a high level of difficulty and Good-counters finish the arithmetic task with no difficulty. 10/36 subjects are in group-B, and 26/36 subjects are in group-G [10]. Recorded EEG of one Good-counter and one Bad-counter sample signal is shown in Fig 1. Those signals can't be recognizable before completing signal augmentation, channel selection and classification process.

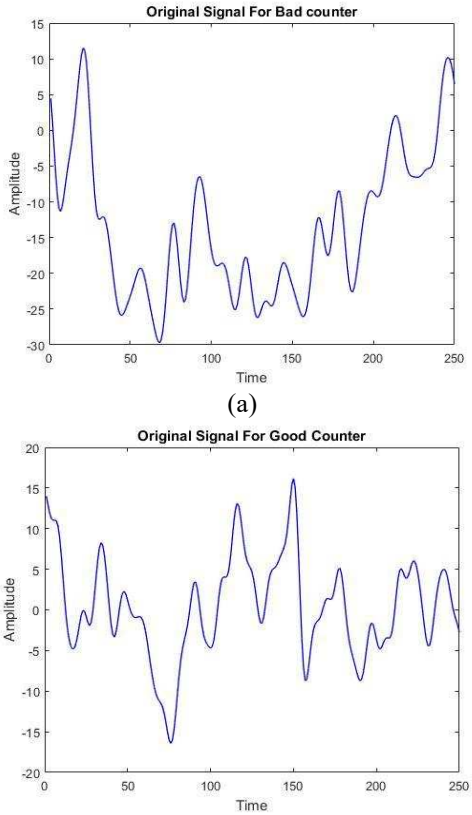

Fig 1: (a) One Good-counter original signal, (b) One Bad-counter original signal.

III. METHODOLOGY

Our proposed methodology has five steps- A. Data Augmentation, B. Dominant Channel Selection, C. Selection of wavelet transform, D. Statistical model selection, E. Classifier selection. Fig 2 shows the block diagram of our methodology.

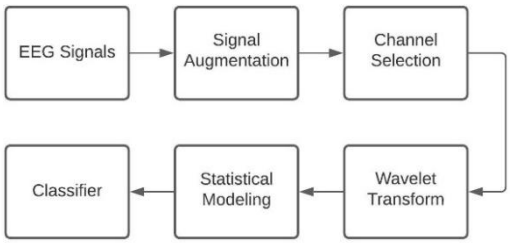

Fig 2: Block diagram of our methodology.

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A. Signal Augmentation

Pre-processing is the basis for any algorithm in signal processing. Our collected EEG data was recorded by a 23 channel Neurocom monopolar device. So, the data has artifact signals. The augmentation process is to remove noise and partitioning data in sub bands. After augmentation we have got EEG sub-bands.

B. Dominant Channel Selection

We have used Human Based Technique for observing all the channel magnitude to select dominant channels. We examine the magnitude of all of our data channels and select five channels for our statistical modeling. We selected those channels which have more than 70% average power magnitude in each signal of the dataset. Fig 3 shows power magnitudes of the channels.

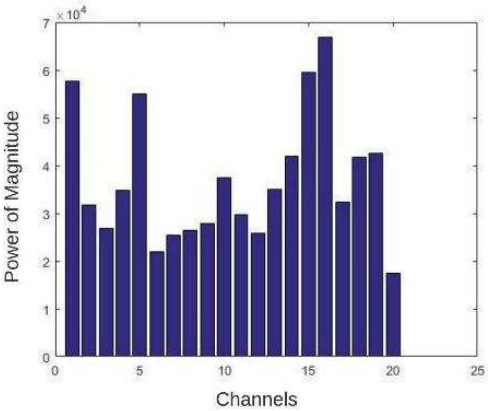

Fig 3: Channels histogram

C. Selection of Wavelet Transform

We have worked on Discrete wavelet transform (DWT) and selected the Haar wavelet for analysis. The Haar wavelet is one of the most basic wavelets and has been employed in a variety of BCI applications. The Haar function can be used to equally estimate any continuous function.

D. Statistical Model Selection

We have chosen the distribution: Gaussian PDF for our data, which best corresponds to the data. Visually we compared some statistical models: Gaussian, t-Location Scale, Stable, Logistic, Extreme and Generalized Extreme model of our EEG signal of complete dataset. Fig 4 shows different statistical model fits.

E. Classifier Selection

We utilized an SVM classifier for our data. For Brain Computer Interface applications, the SVM classifier is commonly used. We also checked the results by using the k-NN classifier. However, SVM is more significant for EEG signals.

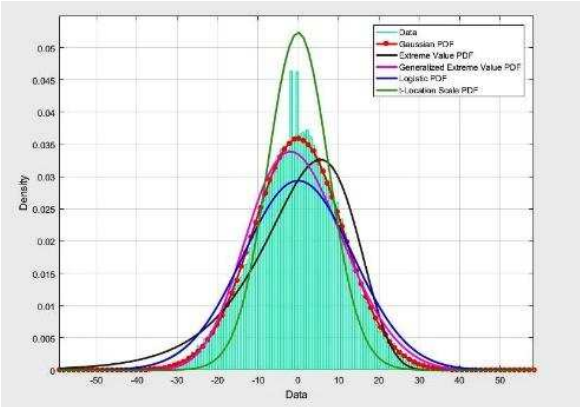

Fig 4: Visualization of some statistical models.

IV. Result analysis

A. Classification Accuracy

We used different classification models and different wavelets and results which are shown in Table I. For assessing our proposed methodology's classification performance results, we have used some known parameters that are standard for evaluating performance- sensitivity, specificity and accuracy. These are the equations for calculating classification parameters.

a) Sensitivity: It is defined as the proportion of appropriate negative labels tagged by our algorithm to all those that are accurate in reality. $\text{Sensitivity}_i = \frac{TP_i}{TP_i+FN_i} \times 100\%$

b) Specificity: Specificity is the algorithm which accurately labels all those who are correctly in reality. $\text{Specificity}_i = \frac{TN_i}{FP_i+TN_i} \times 100\%$

c) Accuracy: It is the proportion of subjects who have been correctly identified to the total number of subjects. Accuracy is the foremost intuitive one. $\text{Accuracy}_i = \frac{TP_i+TN_i}{TP_i+TN_i+FP_i+FN_i} \times 100\%$

Here equations show the terms FP, FN, TP, TN. where FP, FN represent falsely detected positive and negative signals and TP, TN are rightly detected positive and negative signals.

Table I: Classification Accuracies for Statistical modes.

Method	Wavelet	Statistical Model	Sensitivity %	Specificity %	Accuracy %
SVM Classifier	DB4	Logistic	70.8050	22.4359	60.6631
SVM Classifier	DB4	t-Location scale	72.2869	29.4118	70.6541
SVM Classifier	DB4	Gaussian	71.1903	24.1379	60.7975
SVM Classifier	DB3	Gaussian	72.2222	23.1781	72.2222

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SVM Classifier	DB1	Logistic	74.3769	29.5362	49.6864
SVM Classifier	DB1	t-Location scale	73.7233	29.1775	50.6720
SVM Classifier	DB1	Gaussian	74.2258	29.4070	49.5072
SVM Classifier	DB1, DB4	Gaussian	71.9610	27.5113	49.9552

B. Goodness of Features

The Bhattacharyya distance is a measure of divergence. It is successfully used in engineering and statistical sciences. It measures the similarity between two statistical samples. This parameter indicates the intra class compactness. This is a measurement of two group's intra-class compactness. The lower the BD value indicates the class is more compact.

The Geometrical Separability Index used to characterize how separable two statistical samples are. This parameter indicates the inter class distance. If the GSI value is closer to '0' that indicates the classes are inseparable, if the GSI value is closer to '1' that indicates the classes are divisible. The values of BD and GSI of our selected feature are shown in Tables II and III.

C. Comparison with other research

This work presents a statistical modelling of the DWT co-efficient and the classification accuracy is 72.22%. And the process of using this statistical model which was never proposed before for the classification of arithmetic mental tasks. Table IV shows the accuracy comparison between some related works.

Table II: BD Values for the proposed method

Class	Good Group	Bad Group
Values	0.0995	0.0468

Table III: GSI Values for the proposed method

Class	Good-Good	Bad-Bad	Good-Bad
Values	0.0025	0	0.6476

Table IV: Comparison of the classification accuracy with other algorithms.

Sl. No.	Authors	No of Channels	Accuracy
1	B. Ganguly	4	91.67%
2	B. Fatimah	1	98.6%
3	B. Fatimah	1	83.3%
4	Proposed	2	72.22%

CONCLUSION

This study presents the classifications of EEG signals under mental cognitive stress as a Brain Computer Interface work. The dataset was gathered from 36 healthy individuals of similar ages. Here we present a novel methodology to make statistical modelling. We have extracted features using the Support Vector Machine classifier from the given EEG signals of two different groups. Classification results showed that selecting features from all channels and models from all channels resulted in an average accuracy of 72.22%. Accuracy level using this dataset and proposed method satisfy us. Our statistical model-based algorithms had the best accuracy compared to other classification algorithms. This gives researchers a chance to dig deeper into the classification of mental cognitive stress. Many more advanced feature selection and classification methods could be used to further improve the classification of mental arithmetic task performance.

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